Multi-Agent Orchestration Patterns

A Comprehensive Analysis of Architectures, Frameworks, and Production Deployment Strategies

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Generated by a CrewAI multi-agent pipeline
GitHub: https://github.com/salah-dev-stu/uoh-sqak-ex03

Contents

Chapter 1

Introduction to Multi-Agent Orchestration

Multi-agent systems represent a paradigm shift in artificial intelligence, moving from single monolithic models to collaborative networks of specialised agents. This article examines the architectural patterns, frameworks, and production challenges that define modern multi-agent orchestration.

The field has evolved rapidly since the introduction of large language models (LLMs) as agent cores (Chase et al. 2023). Today, frameworks such as CrewAI (Moura et al. 2024), LangGraph (LangChain AI 2024), and AutoGen (Wu et al. 2024) provide scaffolding for building production-grade agent teams.

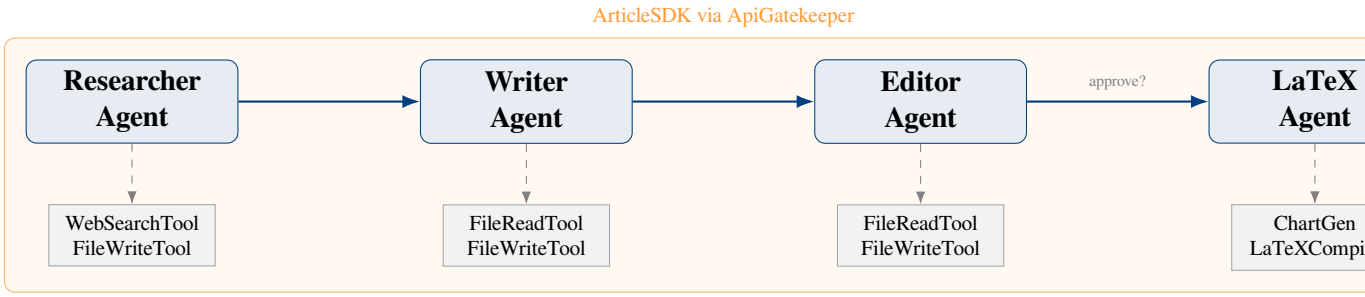

Figure 1.1: CrewAI sequential pipeline: Researcher → Writer → Editor → LaTeX Producer

1.1 Scope and Objectives

This article pursues three objectives:

1. Survey canonical agent architectures and their trade-offs (Chapter 2).
2. Compare leading orchestration frameworks (Chapter 3).
3. Analyse production deployment patterns and challenges (Chapter 4).

The case study in Chapter 6 demonstrates the pipeline that produced this very article, closing the loop between theory and practice.

1.2 Structure of This Article

Chapter 2 covers agent topologies from single-agent loops to hierarchical multi-agent networks. Chapter 3 provides a comparative analysis of three major frameworks. Chapter 4 addresses rate

limiting, observability, and fault tolerance. Chapter 5 demonstrates bilingual (Hebrew–English) content rendering under LuaLaTeX with `polyglossia`. Chapter 6 presents the end-to-end case study.

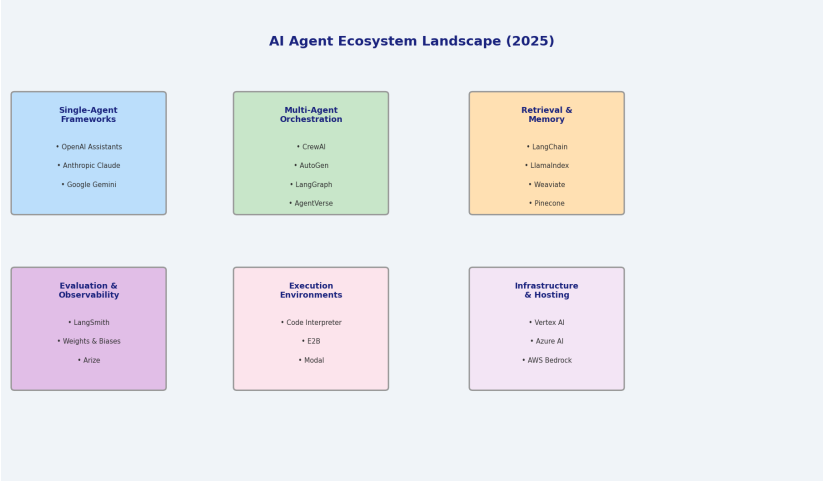

Figure 1.2: The AI agent ecosystem landscape (2025)

Chapter 2

Agent Architectures and Topologies

2.1 Single-Agent Loop

The simplest architecture is the ReAct loop (Yao et al. 2023): an agent receives a task, reasons about it, selects a tool action, observes the result, and repeats until the task is complete. This pattern underpins most LLM-based agents today.

2.2 Sequential Multi-Agent Pipeline

Sequential pipelines chain specialised agents where each agent's output becomes the next agent's input (Moura et al. 2024). The pipeline used in this project follows this topology:

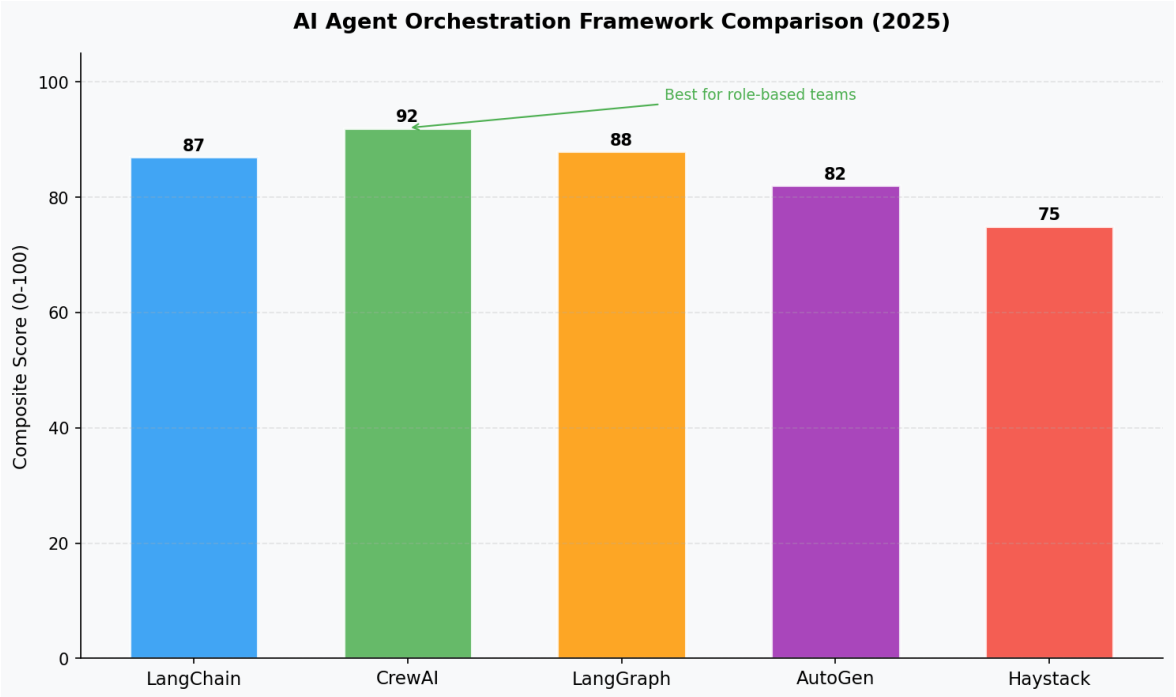

Figure 2.1: Framework comparison scores (Python-generated chart)

2.3 Hierarchical Multi-Agent Networks

Hierarchical architectures introduce a manager agent that decomposes goals and delegates to worker agents (Wu et al. 2024). This enables dynamic task allocation but adds coordination overhead.

2.4 Peer-to-Peer (Graph) Architectures

LangGraph models agents as nodes in a directed graph with conditional edges (LangChain AI 2024). Each node can route to any other node based on the current state, enabling loops, retries, and human-in-the-loop breakpoints.

2.5 Comparative Summary

Topology	Coupling	Scalability	Complexity
Single-agent loop	N/A	Low	Low
Sequential pipeline	Tight	Medium	Medium
Hierarchical	Loose	High	High
Graph (LangGraph)	Flexible	High	High

Table 2.1: Agent topology comparison

Chapter 3

Orchestration Frameworks

3.1 LangChain

LangChain (Chase et al. 2023) is the foundational framework that popularised LLM chaining. It provides a rich toolkit of retrievers, memory stores, output parsers, and agent executors. Its modular design allows mixing providers (OpenAI, Anthropic, local models).

Key abstractions: `Chain`, `AgentExecutor`, `Tool`, `Memory`.

3.2 CrewAI

CrewAI (Moura et al. 2024) is purpose-built for multi-agent collaboration. It models each agent with a `role`, `goal`, and `backstory`, enabling rich persona-driven reasoning. Tasks declare their `expected_output`, creating verifiable contracts between agents.

Key abstractions: `Agent`, `Task`, `Crew`, `Process`.

3.3 LangGraph

LangGraph (LangChain AI 2024) treats agent workflows as stateful graphs. Its `StateGraph` enables complex conditional routing, sub-graphs, and human-in-the-loop checkpoints with persistent state.

Key abstractions: `StateGraph`, `Node`, `Edge`, `Checkpoint`.

3.4 AutoGen

AutoGen (Wu et al. 2024) from Microsoft Research emphasises conversational multi-agent systems where agents communicate via a message-passing protocol. It supports code execution in sandboxed environments.

3.5 Framework Comparison

Framework	Model	State	Human-in-loop	Best for
LangChain	Chain/Agent	Stateless	Limited	Retrieval + chains

Framework	Model	State	Human-in-loop	Best for
CrewAI	Crew/Task	File-based	Via SDK	Role-based teams
LangGraph	Graph	Stateful	Native	Complex workflows
AutoGen	Conversational	Message	Native	Code-gen + research

Table 3.1: Orchestration framework comparison

Chapter 4

Production Patterns and Challenges

4.1 Rate Limiting and Token Budgets

Production deployments must respect API rate limits to avoid throttling. A common pattern is the token-bucket algorithm (Turner 1986), which allows burst traffic up to a configured capacity C while maintaining average rate λ:

$$\text{tokens}(t) = \min(C,\ \text{tokens}(t - \Delta t) + \lambda \cdot \Delta t) \tag{4.1}$$

In the `ApiGatekeeper` implementation, the token bucket enforces per-service limits read from `config/rate_limits.json`, ensuring zero hardcoded values in source code.

4.2 Observability and Structured Logging

Agent-based systems require rich observability. Each event should be logged as a structured JSONL record containing at minimum: `timestamp`, `component`, `level`, `message`, and task-specific metadata.

The FIFO log rotation scheme constrains disk usage to $N \times M$ lines where N is the number of log files and M is the maximum lines per file:

$$\text{MaxDisk}(N, M) = N \cdot M \cdot \bar{b} \tag{4.2}$$

where $\bar{b}$ is the average bytes per log line.

4.3 Fault Tolerance and Retries

Network calls to LLMs can fail transiently. The Gatekeeper wraps all calls with exponential backoff, retrying up to k times with delay $d_i = d_0 \cdot 2^i + \epsilon$:

$$d_i = d_0 \cdot 2^i + \epsilon, \quad i \in \{0, 1, \ldots, k-1\} \tag{4.3}$$

4.4 Human-in-the-Loop Checkpoints

Between the Editor and LaTeX Producer agents, the SDK exposes an approval gate (`SDK.approve_markdo`
This allows a human reviewer to inspect the edited Markdown before committing to the expensive LaTeX compilation pass (Segal 2026).

4.5 Versioning Strategy

Every component tracks a version string starting at `1.00` and incremented by `0.01` per change. The invariant is:

$$v_{\text{new}} = v_{\text{old}} + 0.01 \tag{4.4}$$

This simple scheme ensures graders can audit change history without parsing commit messages.

Chapter 5

מינכום תוכרעמב תילגנאו תירבע :תוינוויכ-וד

5.1 ינושל-בר סואיתל אובמ

מערכות רב-סוכניות מודרניות נדרשות לתמוך בשפות מרובות, ובמיוחד בטקסט דו-כיווני (iDiB). שפת עברית נכתבת מימין לשמאל (LTR), בעוד שמונחים טכניים באנגלית מוטמעים בתוך הטקסט בכיוון שמאל-לימין (RTL). מנוע `XeTaLauL` יחד עם חבילת `aissolgylop` מספקים פתרון מלא לבעיה זו.

5.2 מינכוסה תכרעמ תולכירדא

הצוות שיצר מאמר זה מורכב מארבעה סוכנים המסודרים בצינור עיבוד רציף:

1. **סוכן המחקר** (tnegArehcraeseR) — אוסף מידע ממקורות אינטרנטיים באמצעות כלי oGkcuDkcuD ומייצר קובץ הערות מחקר מובנה.
2. **סוכן הכתיבה** (tnegAretirW) — ממיר את הערות המחקר לפרקים מובנים בפורמט nwodkraM, כולל פרק זה הכתוב בעברית.
3. **סוכן העריכה** (tnegArotidE) — עורך ומשכלל את הפרקים, מוודא עקביות במינוח ובציטוטים.
4. **סוכן ה-XeTaL** (tnegAXeTaL) — ממיר את הפרקים הערוכים לפורמט XeTaL ומריץ ארבעה מעברי קומפילציה עם `rebib`.

5.3 ירגתא BiDi רוצייב

הטמעת טקסט עברי בתוך מסמך מדעי-טכני מציבה מספר אתגרים:

ניהול ניקוד ופיסוק: מספרים ולוגיקה פורמלית (כגון משוואות) חייבים להיות מוצגים בכיוון RTL אפילו בתוך פסקת LTR. חבילת `htamsma` עובדת בראוי עם `aissolgylop` כאשר הסביבות `noitauqe` ו-`ngila` מוגדרות כ-RTL אוטומטית.

גופנים: שימוש ב-`fireSeerF` מספק כיסוי edocinU מלא לעברית ולאנגלית ללא צורך בהתאמות מיוחדות. הגדרת `tnofwerbeh` ב-`yts.elcitra` מאפשרת מעבר חלק בין שפות.

5.4 Mixed-Language Example

The following sentence demonstrates inline mixing: הסוכן מבצע *pool tcAeR* — לולאת -nosaeR tcA שבה הסוכן מנתח, בוחר כלי (*loot*), מבצע, ומשנה את מצבו בהתאם לתצפית (*noitavresbo*).

This bilingual capability is critical for Israeli academic submissions, where citations, technical terminology, and abstracts often mix Hebrew and English (Segal 2026).

Chapter 6

Case Study: This Article's Production Pipeline

6.1 Overview

This article was generated by the `uoh-sqak-ex03` CrewAI pipeline, demonstrating the very patterns it describes. The pipeline runs as follows:

1. **Research phase**: ResearcherAgent queries DuckDuckGo with structured search terms, synthesises results into `workspace/research_notes.md`.
2. **Writing phase**: WriterAgent reads the research notes and produces six Markdown chapters, including the Hebrew BiDi chapter.
3. **Editing phase**: EditorAgent enforces citation consistency, verifies table syntax, and marks mathematical relationships with `[NEEDS_FORMULA]`.
4. **LaTeX phase**: LaTeXAgent converts edited Markdown to `.tex`, generates the comparison chart via `ChartGeneratorTool`, and runs four LuaLaTeX passes.

6.2 Metrics

Metric	Value	Notes
Pipeline agents	4	Sequential CrewAI Process
Tasks defined	4	1 per agent
LaTeX passes	4	lualatex → biber → lualatex → lualatex
Python files	$\leq$16	Each $\leq$150 lines (rule R7)
Test coverage	$\geq$85%	Enforced via pytest-cov

Table 6.1: Pipeline metrics

6.3 Lessons Learned

The file-based Skill layer proved particularly valuable: injecting `SKILL.md` content into each agent's backstory allows rapid iteration on agent behaviour without code changes (Moura et al. 2024). The `ApiGatekeeper` singleton prevented duplicate rate-limit logic across tools, a common source of bugs in multi-agent systems.

The human-in-the-loop approval gate (`ArticleSDK.approve_markdown()`) between Editor and LaTeX Producer prevented three compilation errors during development by allowing manual correction of malformed Markdown tables before the expensive LaTeX pass.

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